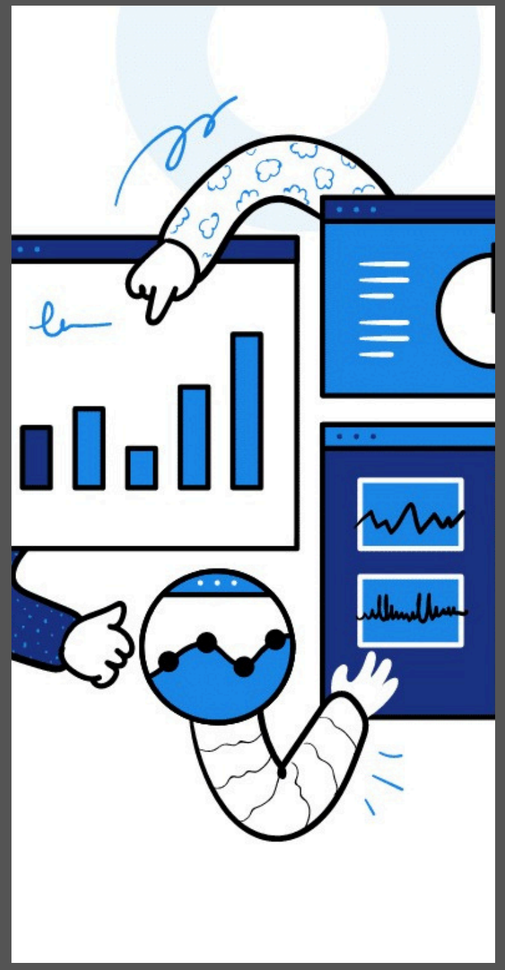
Inside The Story of Why Employees Leave

To transform employee data into predictive insights about workforce attrition



By-
Peddi Praharshita
peddipraharshita14828@gmail.com
peddipraharshita18@gmail.com

what is Employee Attrition?

Thesteady outflowofemployeesfrom an organization overtime.

Why does it Matter?

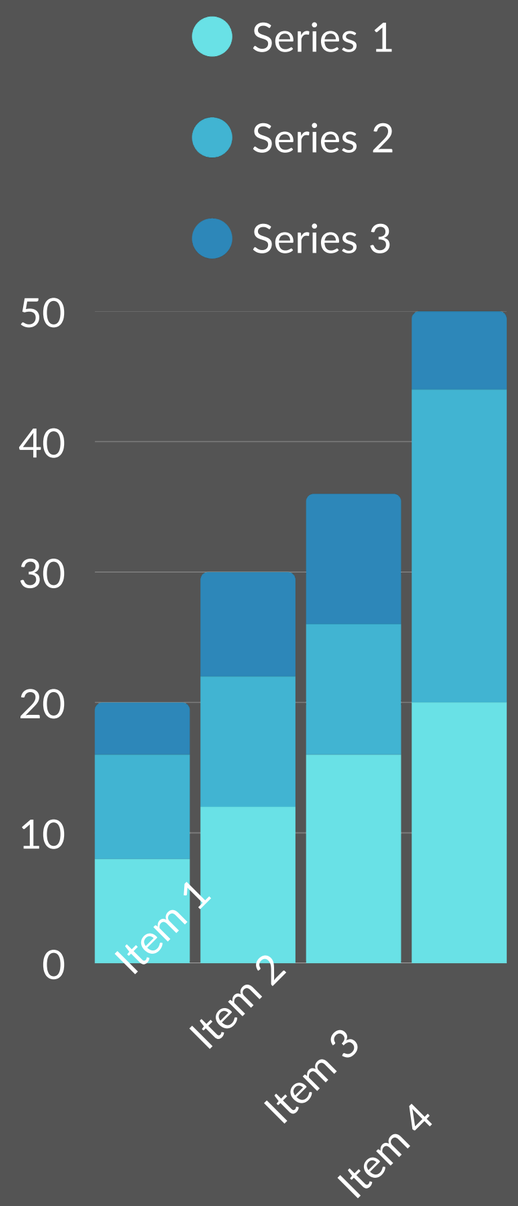
Teamslose experience, productivity drops, and companies spendmore replacing talent.

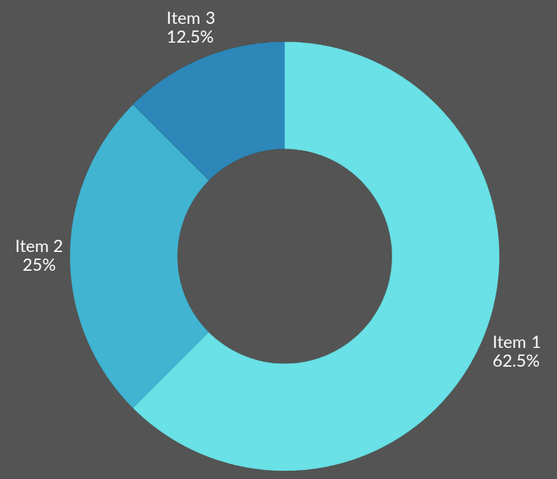
Our Ideation

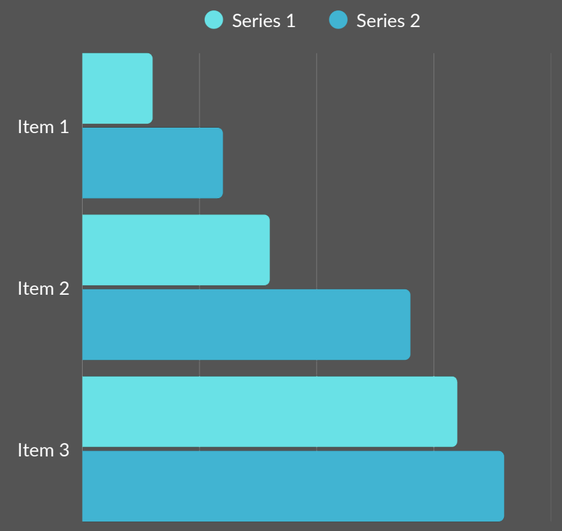
By examining HRdata across roles, demographics, and work patterns, we can discover what leads employees to exit

Project Aim

Touncoverthe most influential factors contributing to attrition using Power BI analytics and machine learning feature insights.







Dataset Used

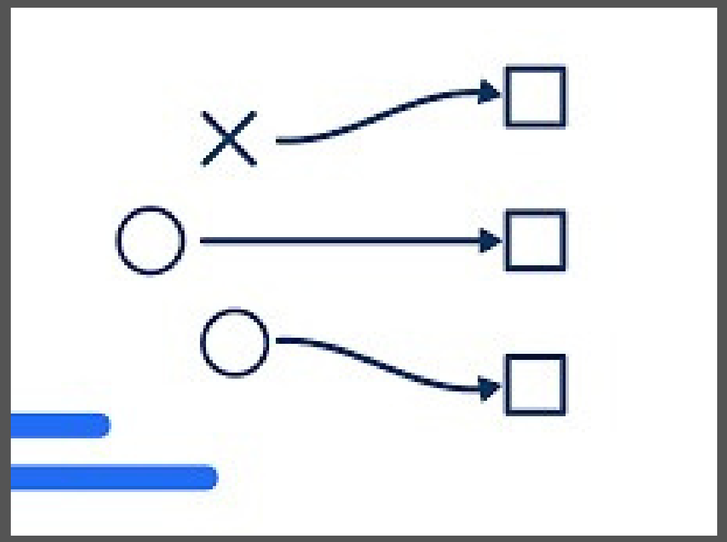
IBM HR Employee Attrition Dataset

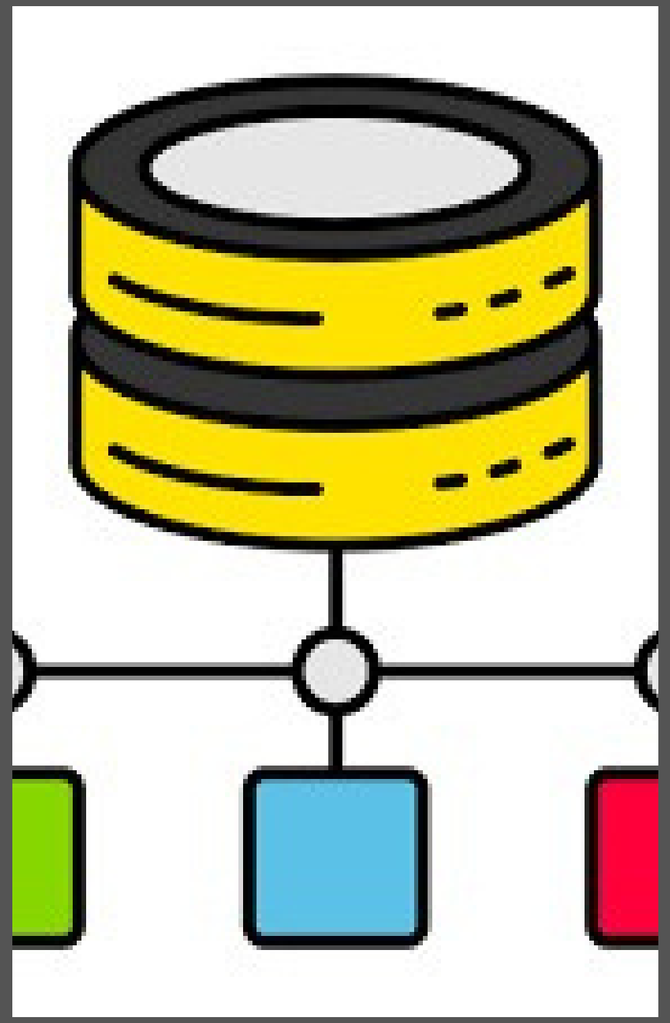
Dataset Size

- 1470employee records
- 35variables covering demographics, job details, performance, and separation status

Purpose

To examine how different employee characteristics relate to attrition and identify which attributes influence exits the most.





Data Categories Included

- Personal:Age,Gender, Marital Status
- JobDetails: Role, Department, Travel, Job Level
- Experience: Years at Company, Years in Current Role, Promotions
- Work Conditions: Distance, Overtime, Work-life Balance
- Satisfaction Scores: Environment, Relationship, Job Involvement
- Attrition Label: Yes / No

KPIs Summary

1470 Total Employee	16.12% Attrition Rate	15.21 Avg Percent Salary Hike
2.73 Avg Job Involvement	2.72 Avg Environment Satisfaction	

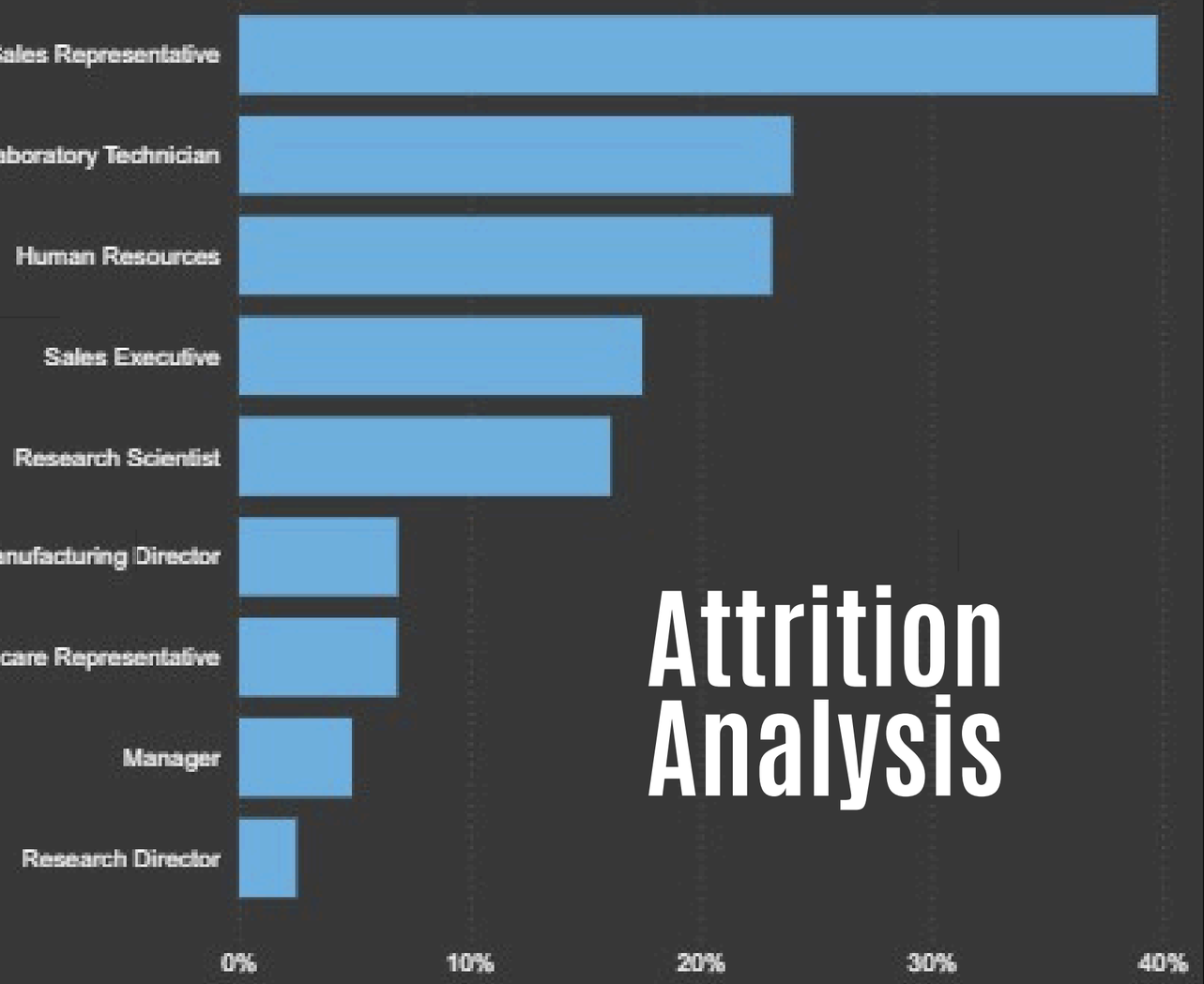
16.12%
Attrition Rate

OverTime
All

EducationField
All

MaritalStatus
All

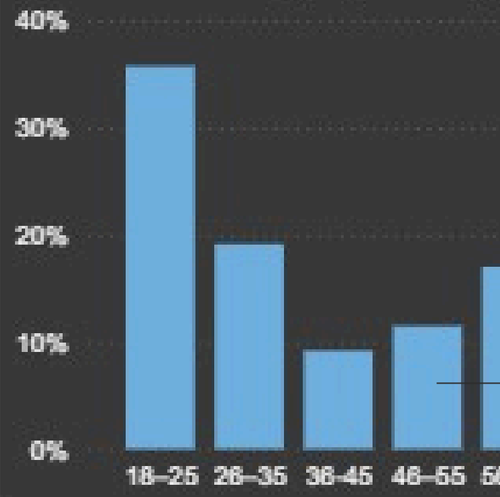
on Rate by Job Role



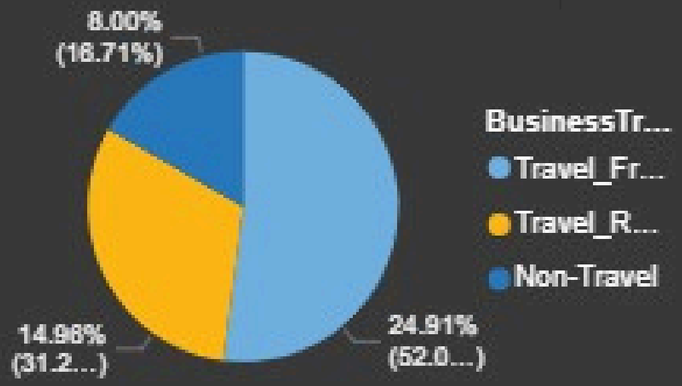
Attrition Analysis

Attrition Rate by Department

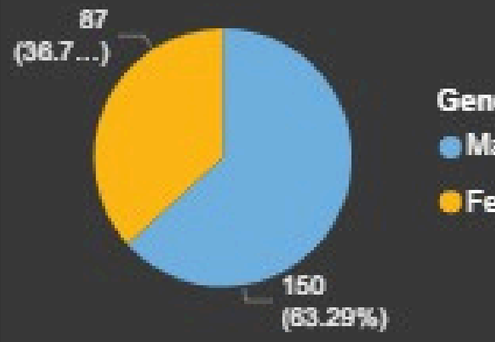
Attrition Rate by Age Group



Attrition Rate by BusinessTravel



Attrition Rate(yes) by Gend



Demographic Analysis

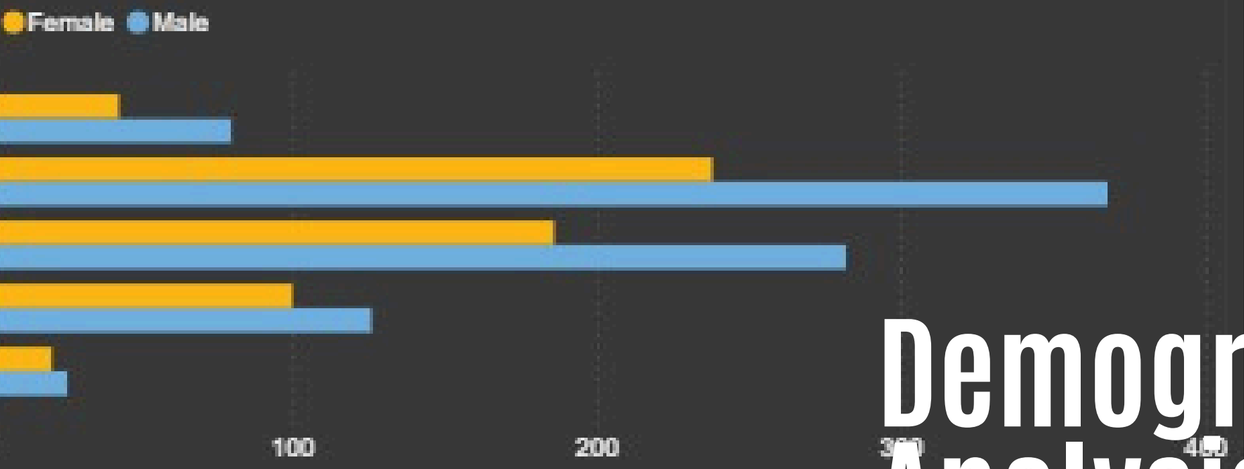
1470
Total Employee

JobRole
All

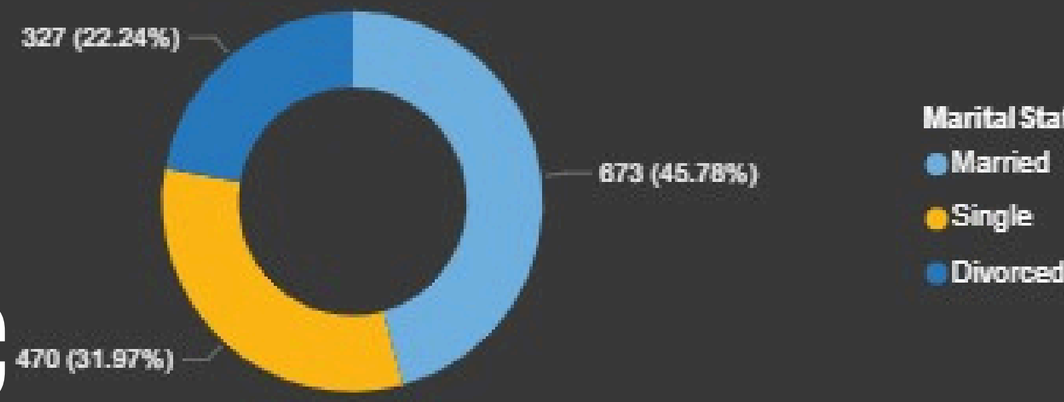
BusinessTravel
All

Distance from Home GroupDistance fro...
All

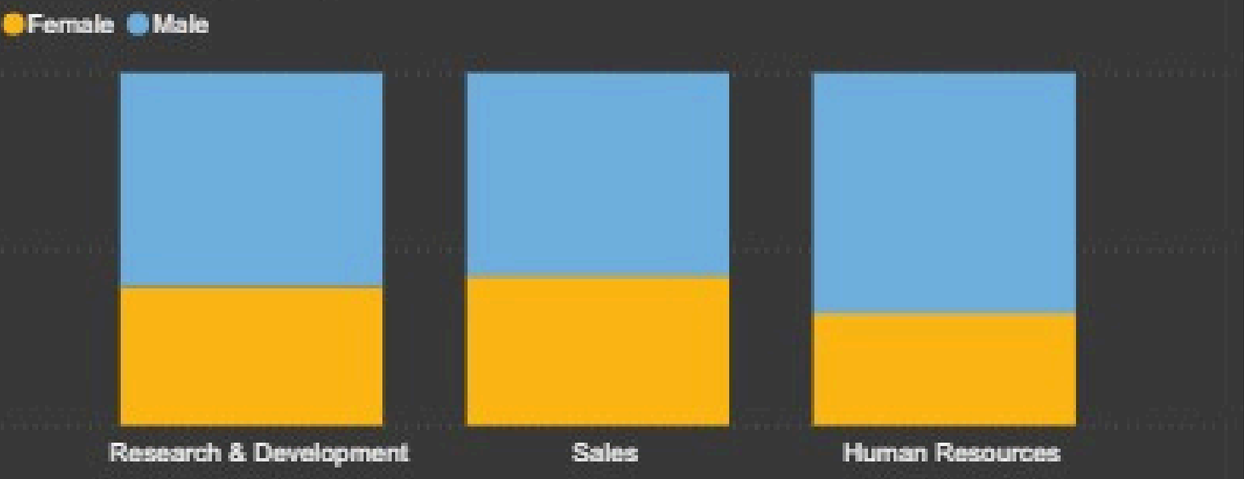
Employee by Age Group and Gender



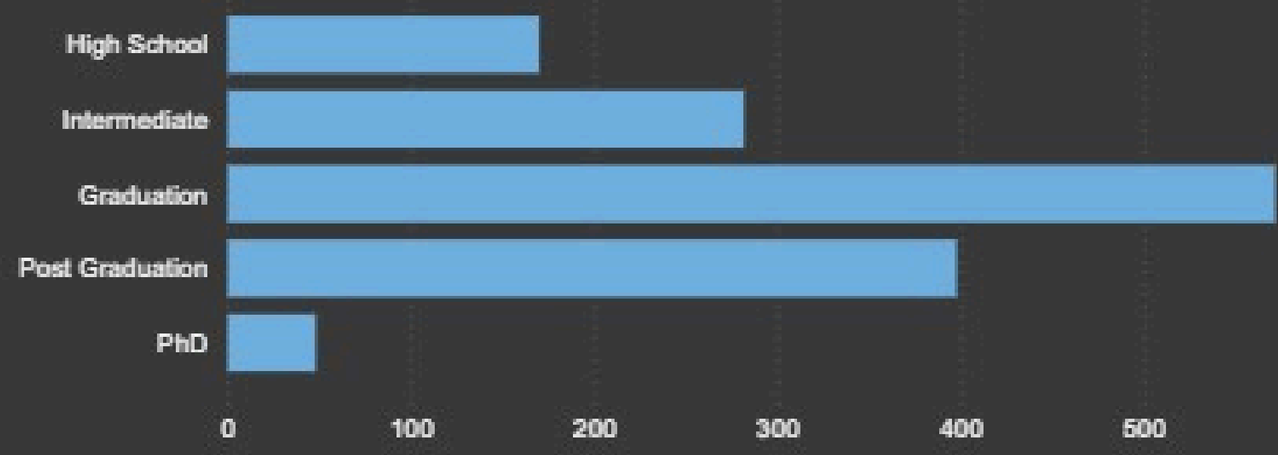
Total Employee by Marital Status



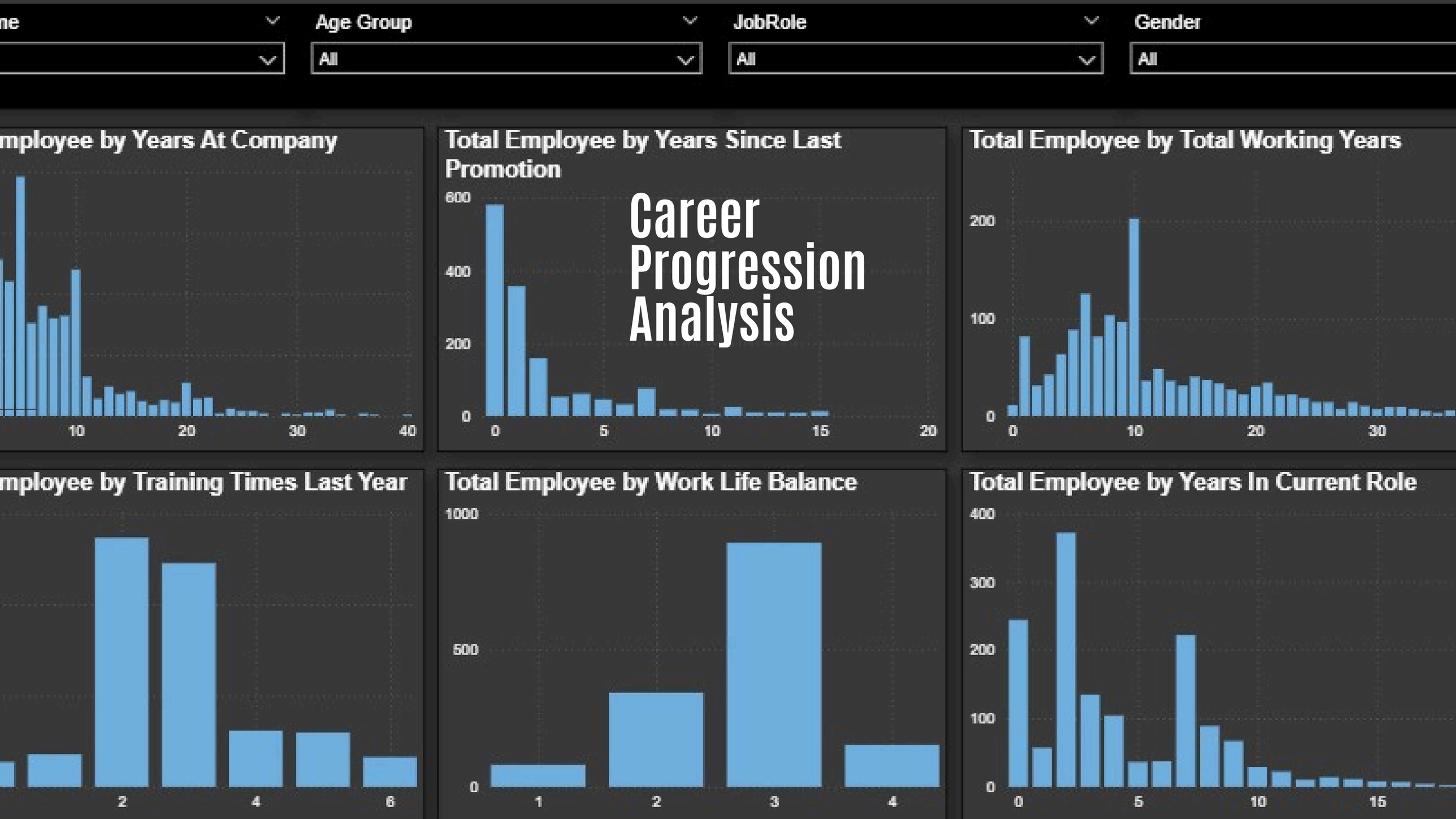
Employee by Department and Gender



Total Employee by Order



Career Progression Analysis

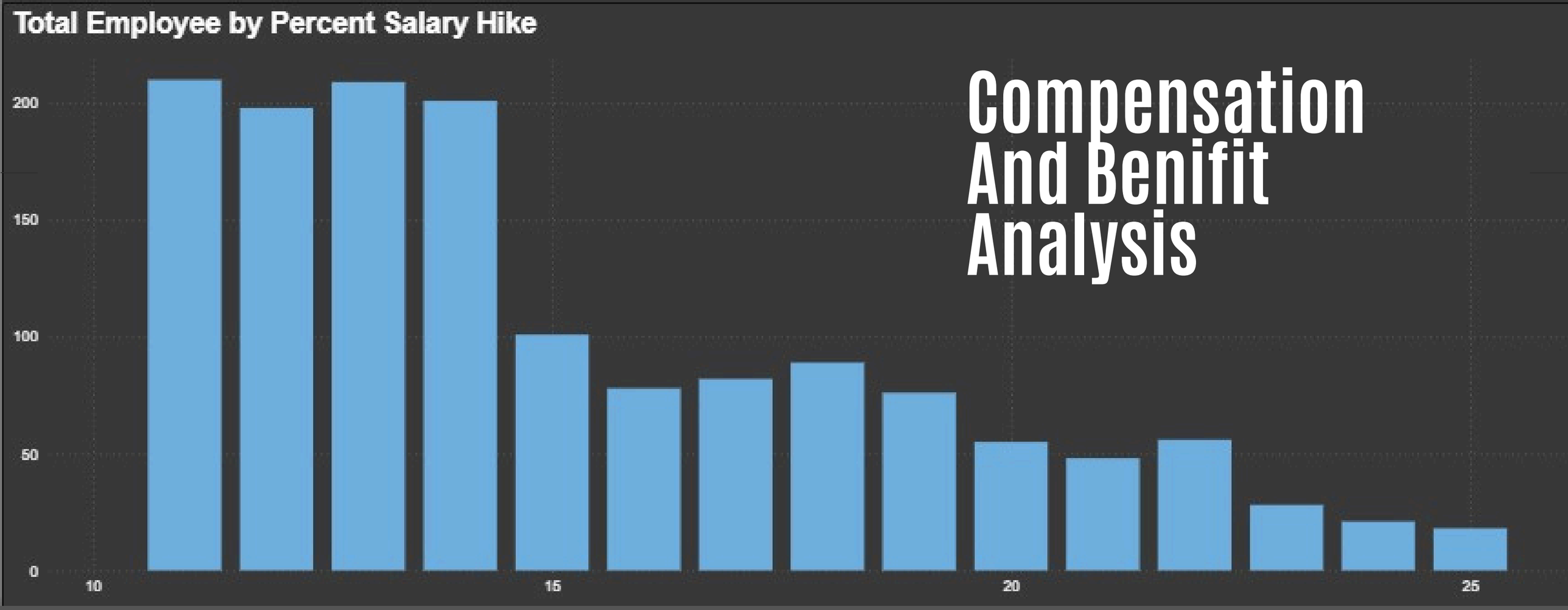


JobRole
All

OverTime
All

EducationField
All

Gender
All



Compensation And Benifit Analysis

Geographical Analysis

JobSatisfaction

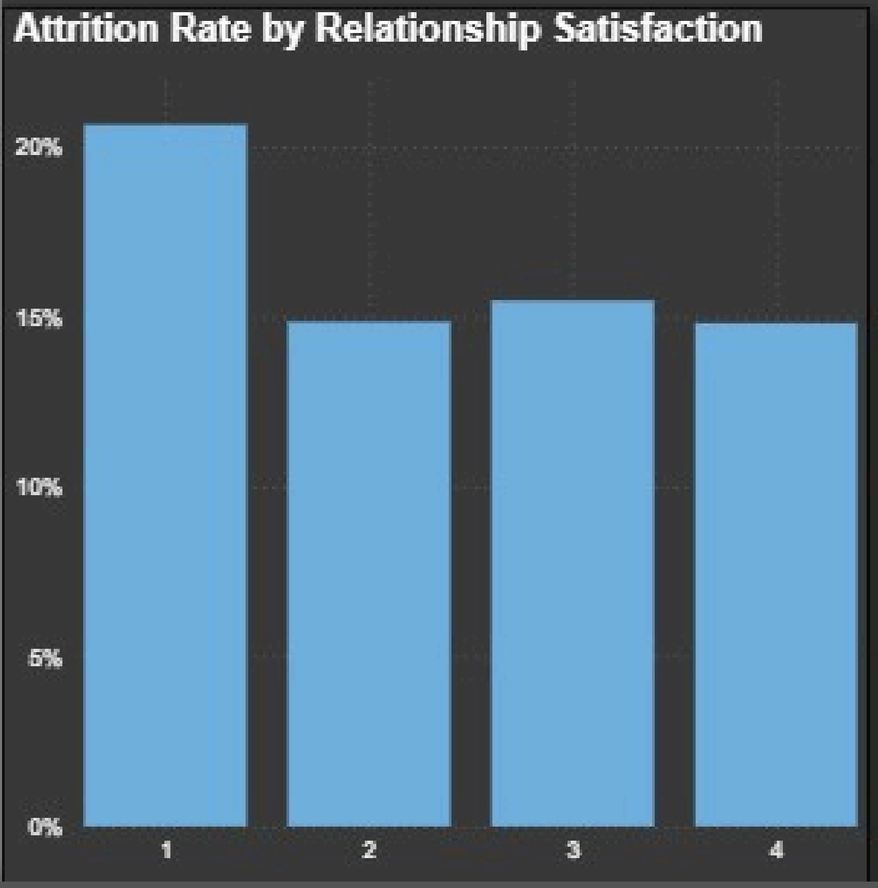
1 4

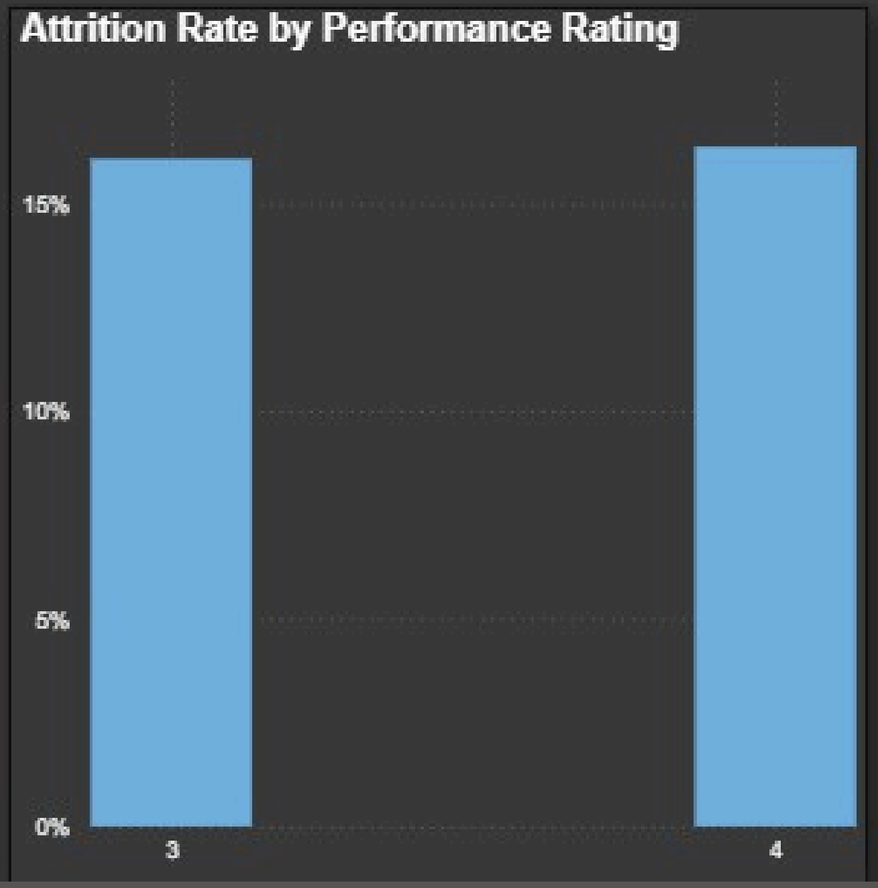
Attrition

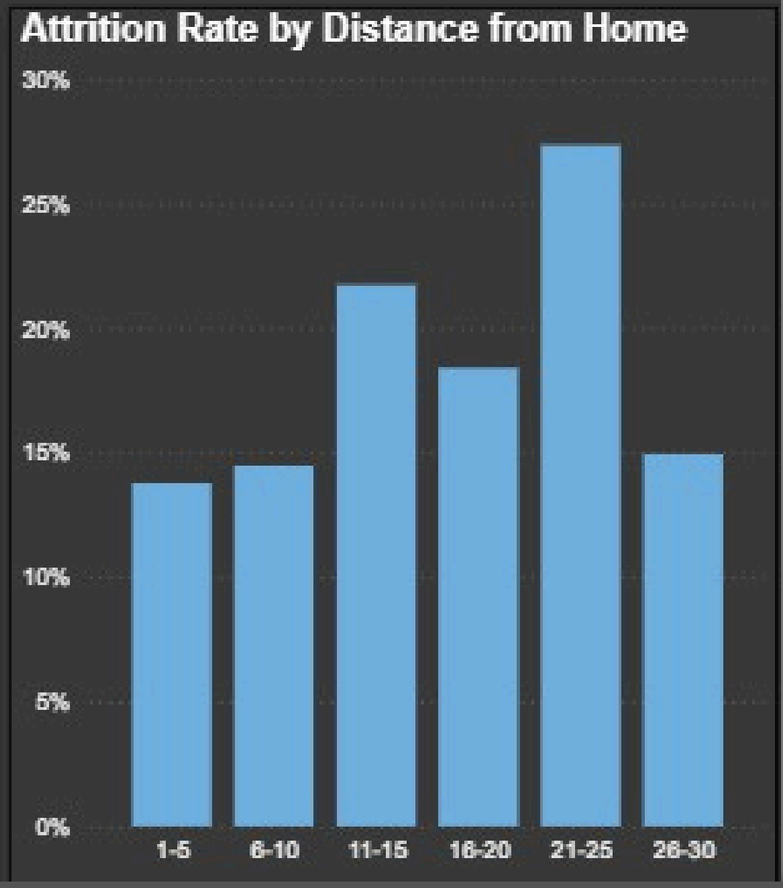
- No
- Yes

Gender

- Female
- Male



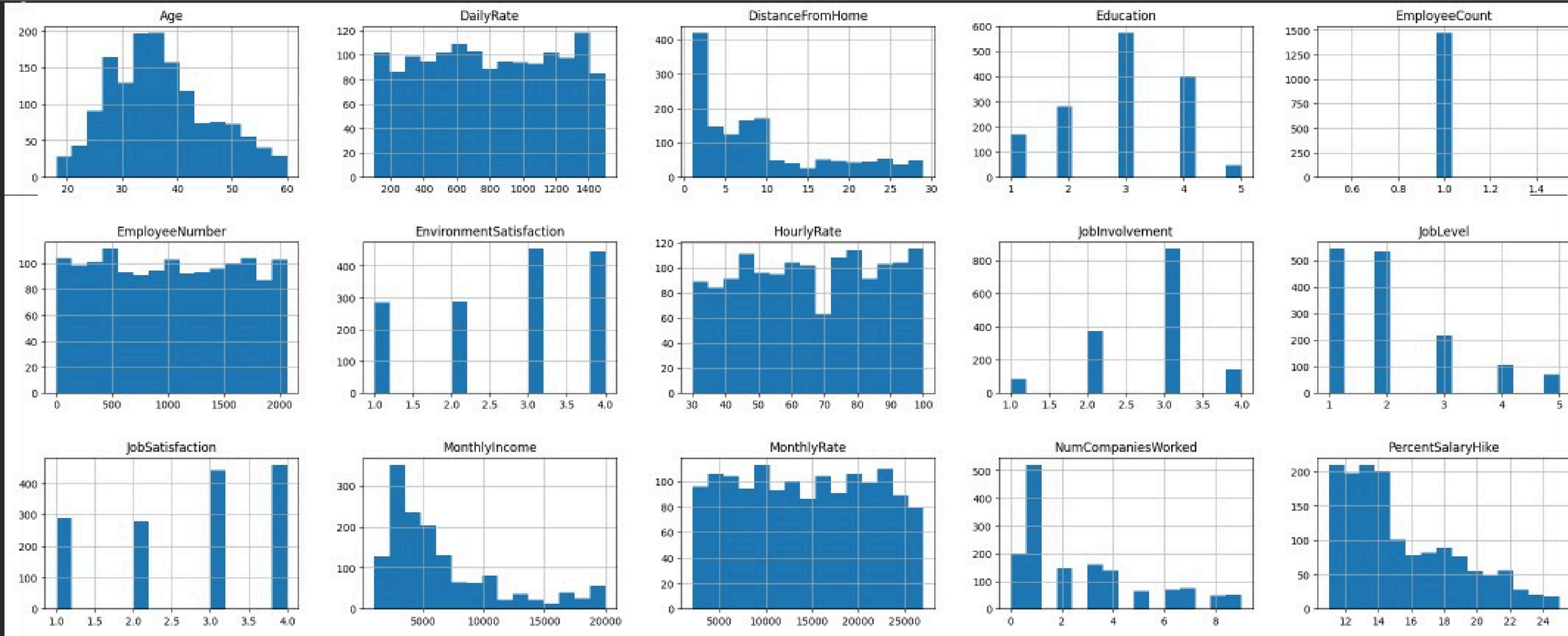


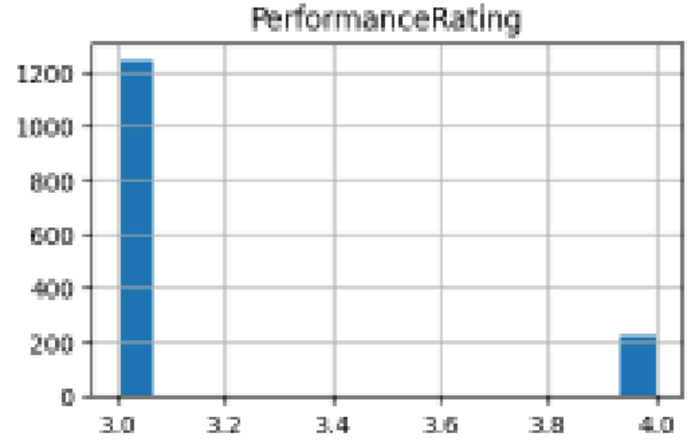


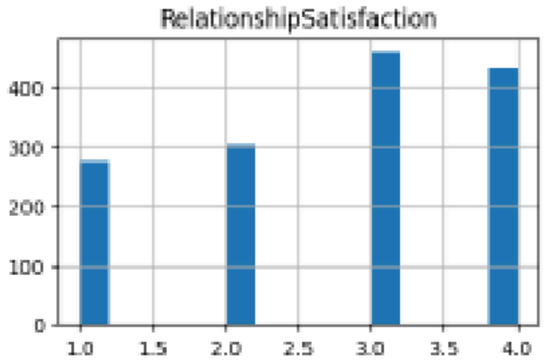
Data Quality Check

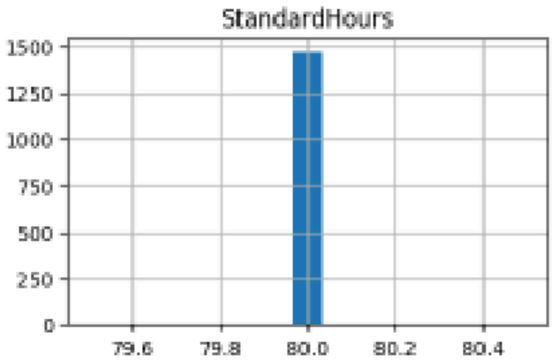
MissingValues : There were no null valuesobserved and outliers were cappedusing Inter Quartile Method

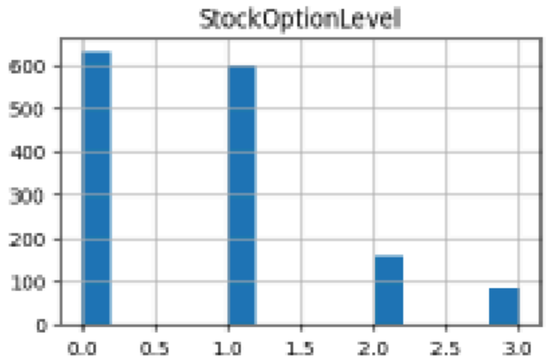
Exploratory Data Analysis

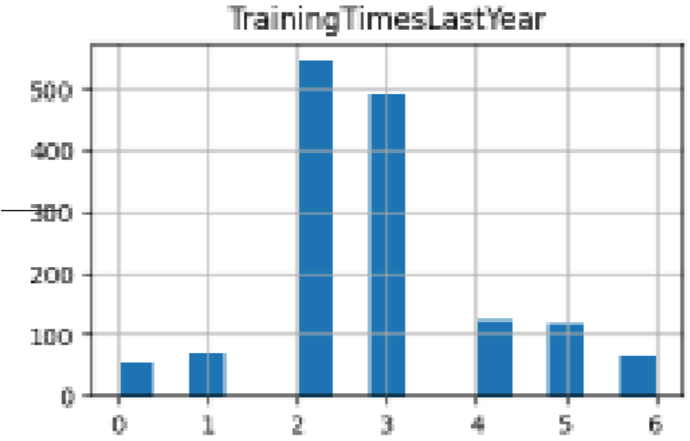



PerformanceRating

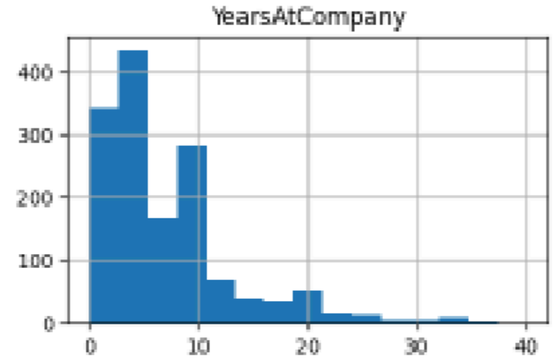

RelationshipSatisfaction

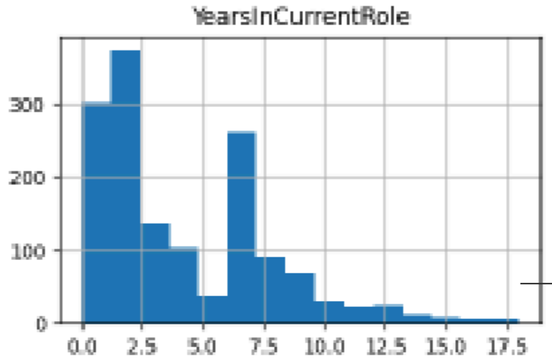

StandardHours

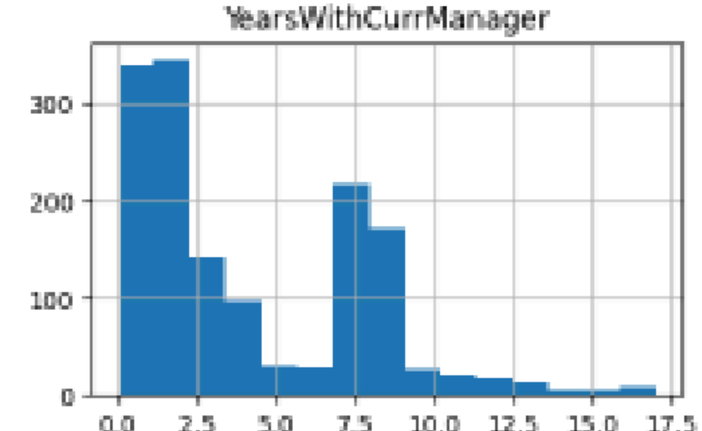

StockOptionLevel

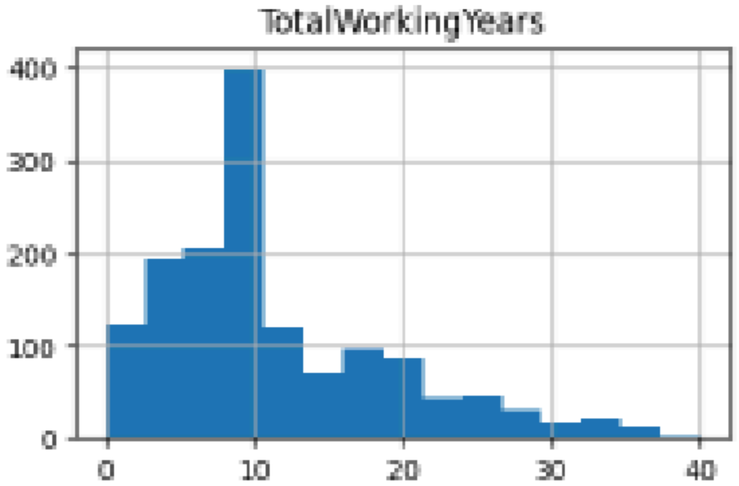

TrainingTimesLastYear

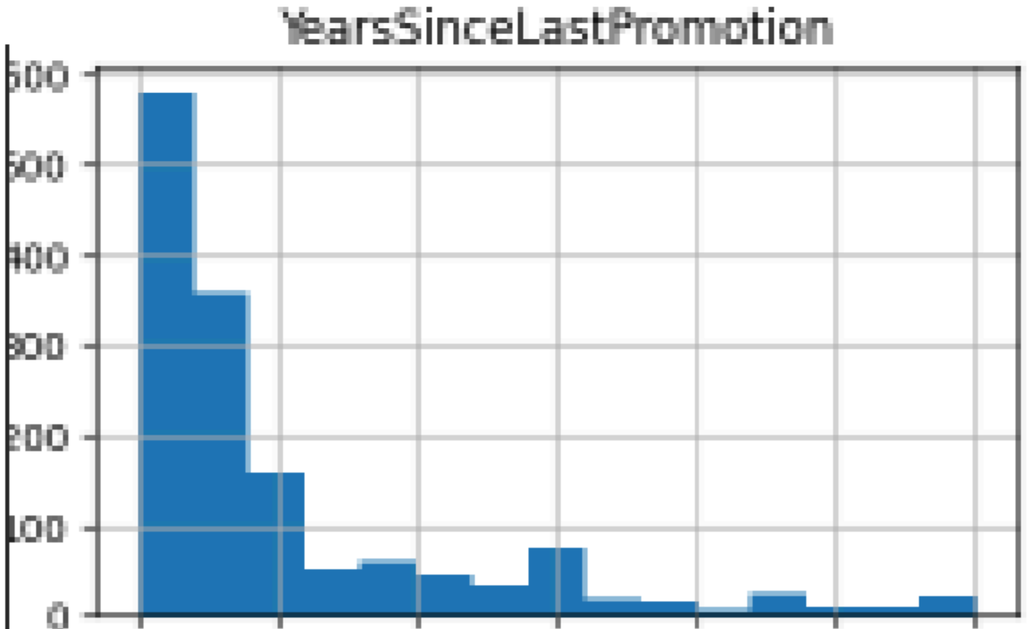
WorkLifeBalance


YearsAtCompany


YearsInCurrentRole


YearsWithCurrManager


TotalWorkingYears


YearsSinceLastPromotion

Right Skewed Distribution

YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager, TotalWorkingYears, DistanceFromHome, NumCompaniesWorked, PercentSalaryHike, MonthlyIncome, StockOptionLevel, TrainingTimesLastYear

Symmetric Distribution

Age, DailyRate, MonthlyRate, HourlyRate

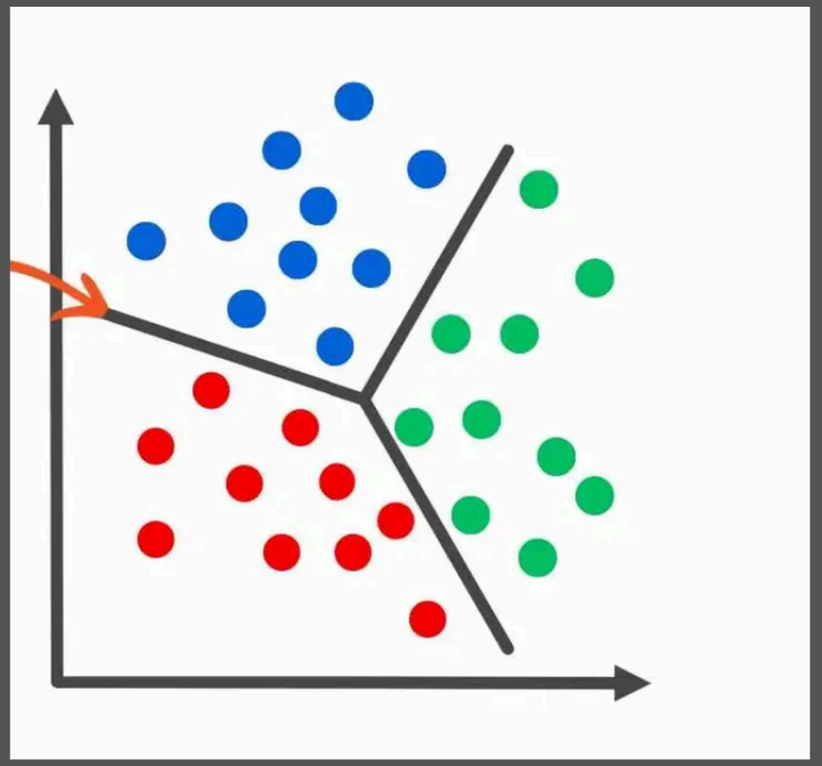
Models Implemented

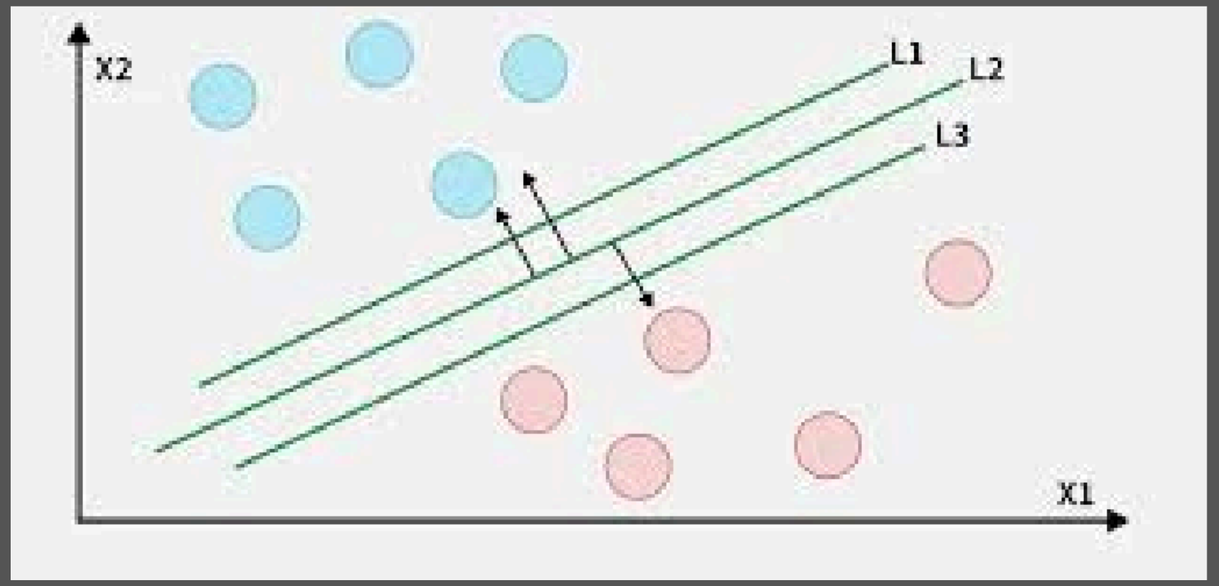
Logistic Regression, CatBoost, XGBoost, Random Forest, Support Vector Machine (SVM)

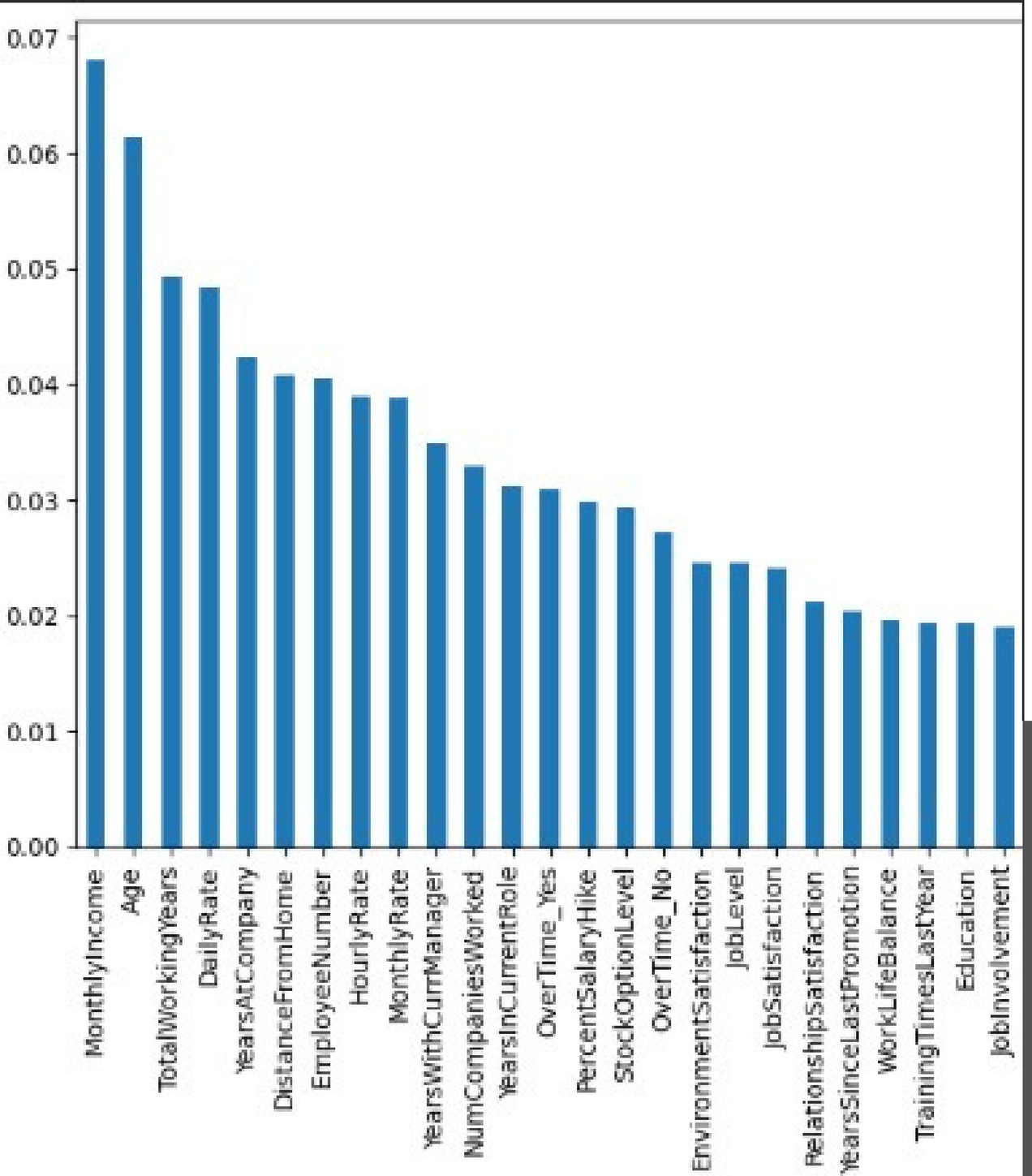
Test Size	Model	Precision	Recall	F1 Score	AUC
0.2	Logistic Regression	0.615385	0.340426	0.438356	0.810664
	Random Forest	0.833333	0.106383	0.188679	0.784434
	SVM	0.451613	0.595745	0.513761	0.806185
	XGBoost	0.769231	0.212766	0.333333	0.793178
	CatBoost	0.608696	0.297872	0.400000	0.792661
0.3	Logistic Regression	0.707317	0.408451	0.517857	0.822383
	Random Forest	0.636364	0.098592	0.170732	0.778493
	SVM	0.494253	0.605634	0.544304	0.807195
	XGBoost	0.640000	0.225352	0.333333	0.764636
	CatBoost	0.606061	0.281690	0.384615	0.759612
0.4	Logistic Regression	0.683333	0.431579	0.529032	0.824362
	Random Forest	0.764706	0.136842	0.232143	0.791673
	SVM	0.500000	0.652632	0.566210	0.815480
	XGBoost	0.729730	0.284211	0.409091	0.800448
	CatBoost	0.738095	0.326316	0.452555	0.804078

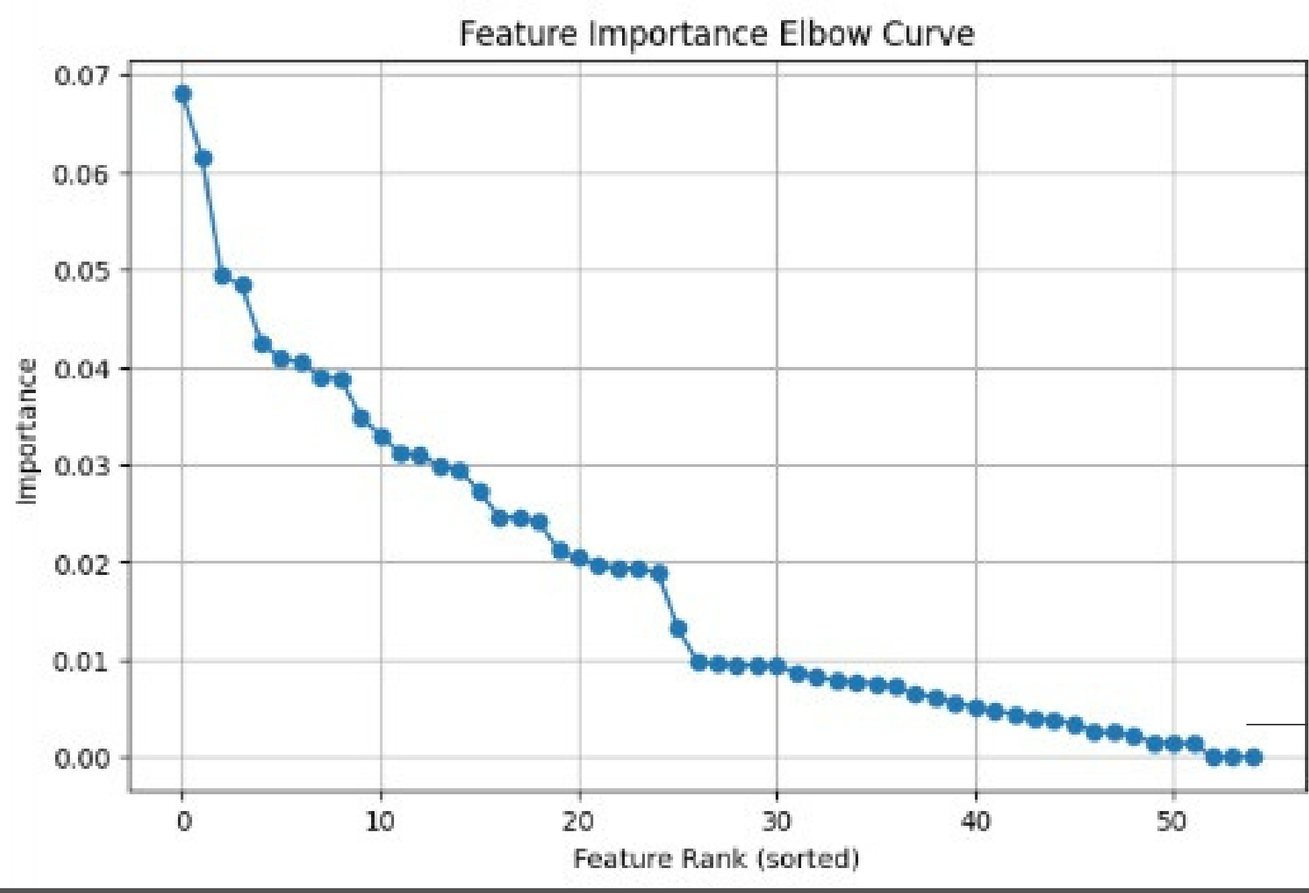
Conclusion Based On the ML Analysis

SVM achieved the highest F1-scores (0.51 - 0.56) across all test splits, showing the strongest balance between precision and recall. This means SVM correctly identifies more employees who truly attrited while keeping false alarms low. Therefore, SVM is the most reliable model for analyzing attrition patterns in this dataset.









- **Top Features which affect the attrition rate mainly determined by the elbow plot on Random Forest**

Why Random Forest?

because it not only performs well on tabular data but also provides featureimportance scores, making it ideal for identifying the key drivers behind employee attrition

Insights (Power BI + ML)

- Salesroles exhibit the highest attrition, driven by heavy workloads and lower job satisfaction.
- Mid-career employees (36–45) are most likely to leave due to limited promotion opportunities and stagnation.
- Employees with <5 years at the company are more vulnerable to quitting, confirming early-tenure risk.
- Low relationship and environment satisfaction strongly predict attrition, highlighting the importance of manager support.
- Long commute distances and overtime hours consistently push employees toward resignation, impacting work-life balance.

Suggestions

- Reduce workload pressure in sales roles through resourcing, automation, or shift balancing.
- Create clear and timely promotion pathways to retain mid-career employees and prevent career stagnation.
- Strengthen onboarding and early employee support to improve retention within the first 2–3 years.
- Improve workplace culture and manager training to boost relationship satisfaction and employee engagement.
- Introduce flexibility options—hybrid work, remote days, or commute allowances—to support work-life balance.

THANK YOU!